Heterophily-aware Adaptive Knowledge Distillation for Hypergraph Neural Networks

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Limitations of MLP

- Although performance of MLP is good when there are sufficient information, in real-world situations, information is usually not enough

You
- Age : ??
- Gender : F

Your Real Age : 21

User A
- Age : 21
- Gender : M

User B
- Age : 21
- Gender : F

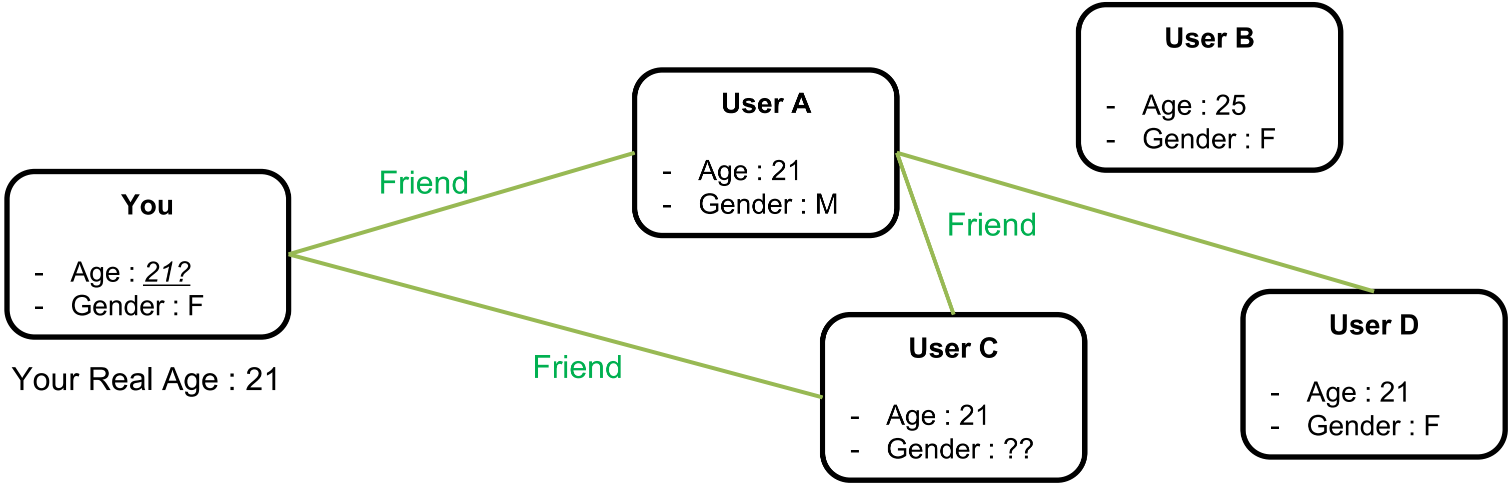
User C
- Age : 21
- Gender : ??

User D
- Age : 25
- Gender : F

System Guessing Your Age

Use of Relation Information by Graphs

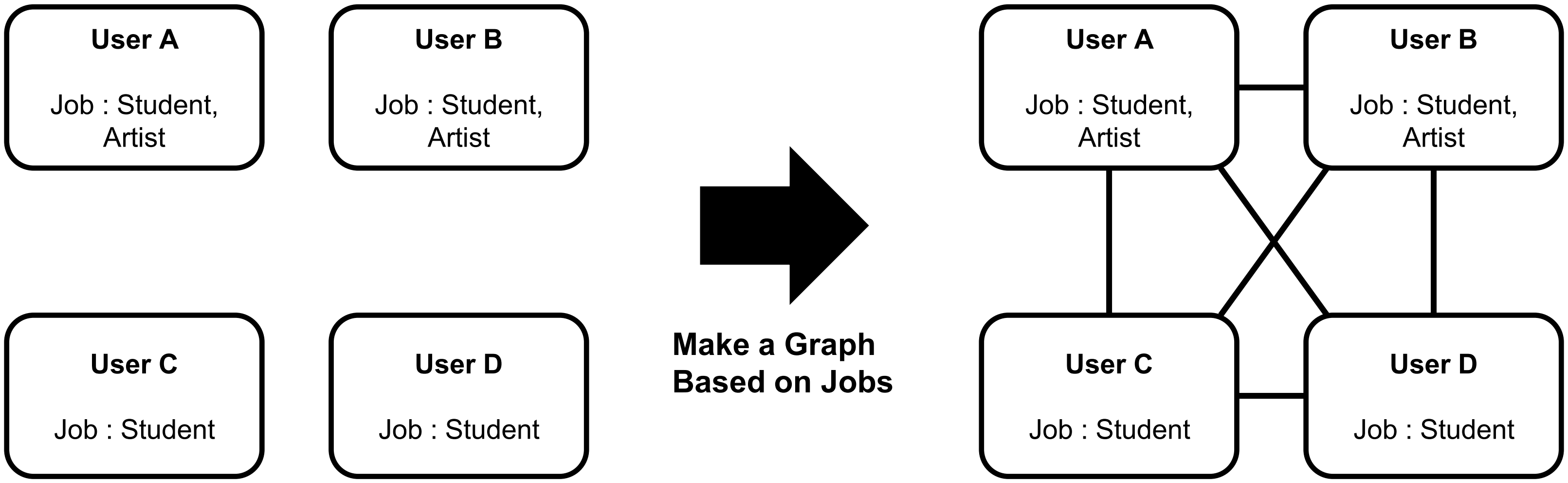
- In real-world, most of the entities involve relationships, which might be useful when the features are not enough for prediction



System Guessing Your Age

Limitation of Simple Graphs

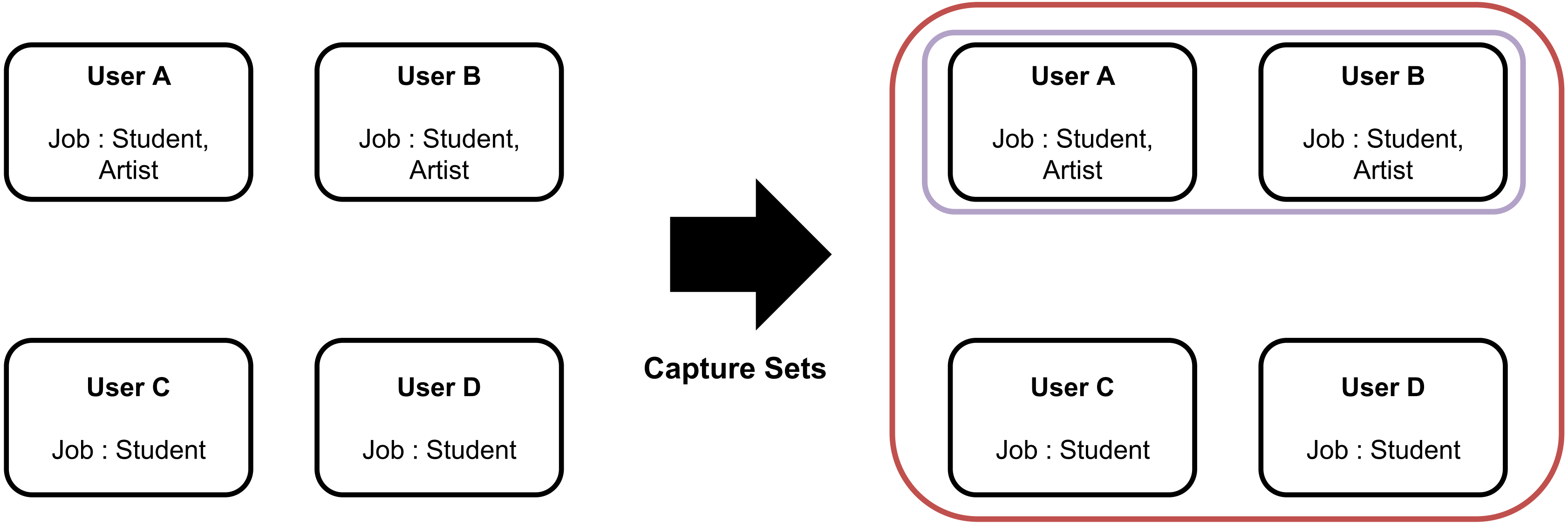
- Graphs are appropriate for representing bipartite relationships; however, it cannot capture set relationships well



Who are the Artists?

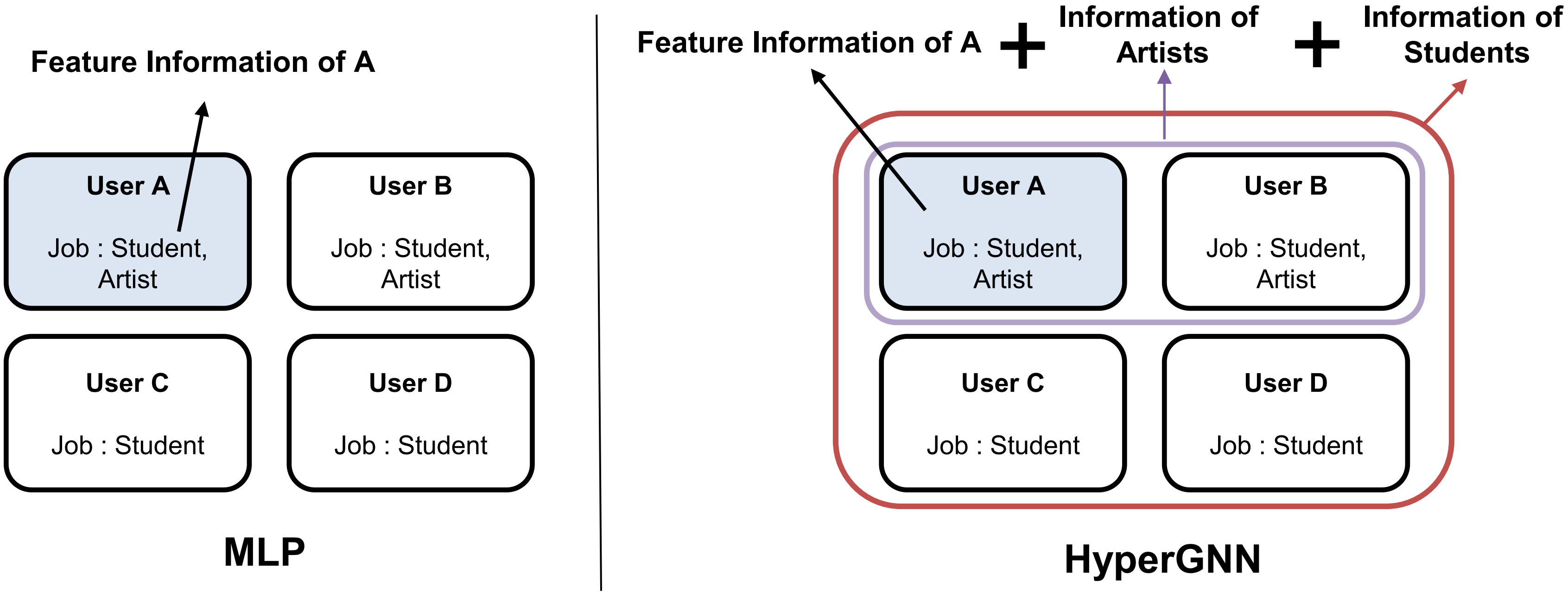
Use of Hypergraphs to Capture Set Information

- By using hypergraphs that can manage the use of set relationships, we can use much more information in real life



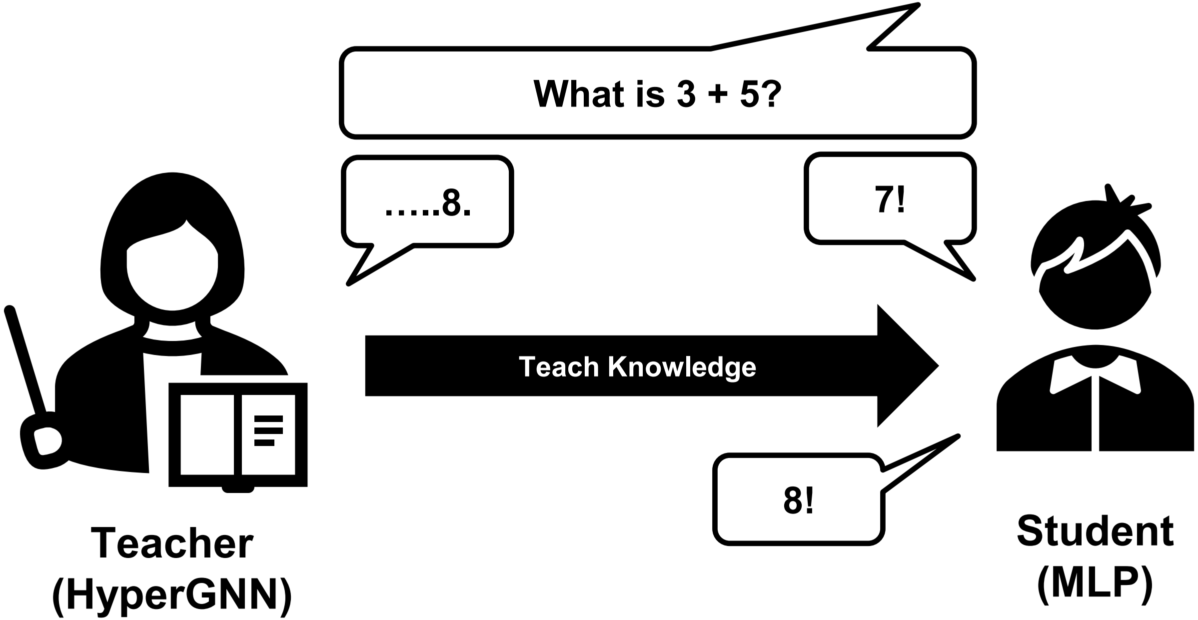
Speed of Hypergraph Neural Network

- Unlike simple MLP, Hypergraph Neural Network (HyperGNN) have to utilize information from the neighbors as well, making it very slow



Use of Distillation to Improve Speed

- If an accurate teacher (HyperGNN) teaches useful knowledge to an inaccurate student (MLP), the final model will become smart and accurate



Is the Teacher Always Better Than The Student?

- Aren’t there situations that MLP is better than HyperGNN?

What is the MBTI of User C? (Answer : ENFP)

Case 1

User A	**User B**	**User C**	**User D**	**User E**
MBTI : ENFP	MBTI : ENFP	MBTI : ????	MBTI : INTJ	MBTI : INTJ
Hobby : SNS	Hobby : Game	Hobby : SNS	Hobby : SNS	Hobby : Rock

HyperGNN : ENFP (Based on Relationship) | MLP : Not Sure (Based on Hobby)

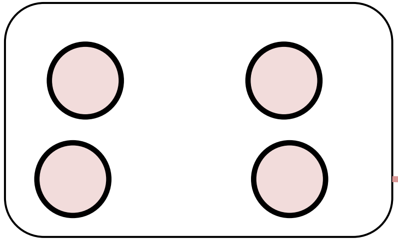
Case 2

User A	**User B**	**User C**	**User D**	**User E**
MBTI : ENFP	MBTI : ISTJ	MBTI : ????	MBTI : ENFP	MBTI : INTJ
Hobby : SNS	Hobby : Game	Hobby : SNS	Hobby : SNS	Hobby : Rock

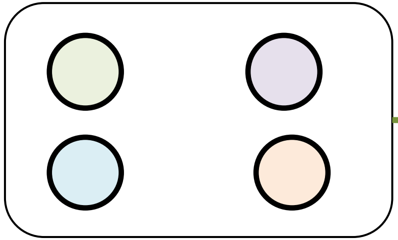
HyperGNN : Not Sure (Based on Relationship) | MLP : ENFP (Based on Hobby)

Heterophily of the Hypergraphs

- If the labels of a hyperedge are the same, it is called a homophily graph
- If the labels of a hyperedge are diverse, it is called a heterophily graph
- How high the hyperedge heterophily is, can be computed by <u>entropy function</u>



Homophily Graph



Heterophily Graph

Entropy Function

$$h(e) = -\sum_{c \in C} p_c \log p_c$$

$\boldsymbol{e}$: *Hyperedge*

C : *Set of Class Labels*

$\boldsymbol{p_c}$: *Proportion of nodes in hyperedge e belonging to class c*

Entropy Low

Entropy High

Heterophily of the Nodes

- Use entropy of label distributions to retrieve hyperedge heterophily scores, and then average it for hypernode heterophily

Hyperedge Heterophily

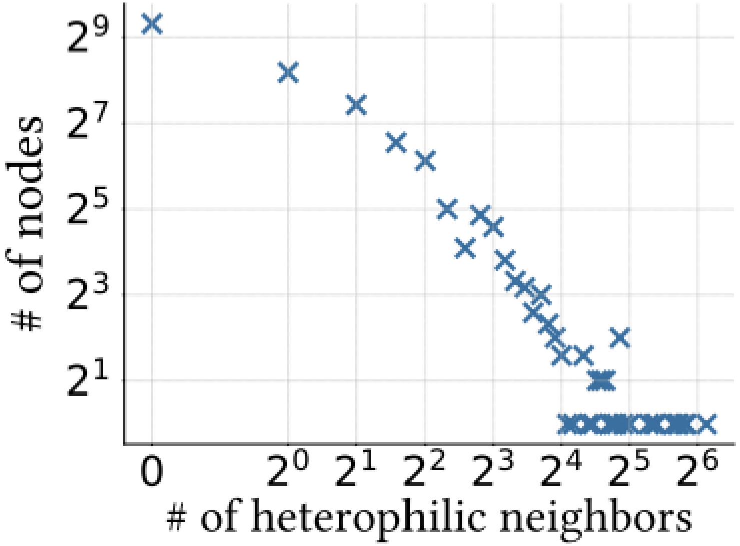
$$h(e) = -\sum_{c \in C} p_c \log p_c,$$

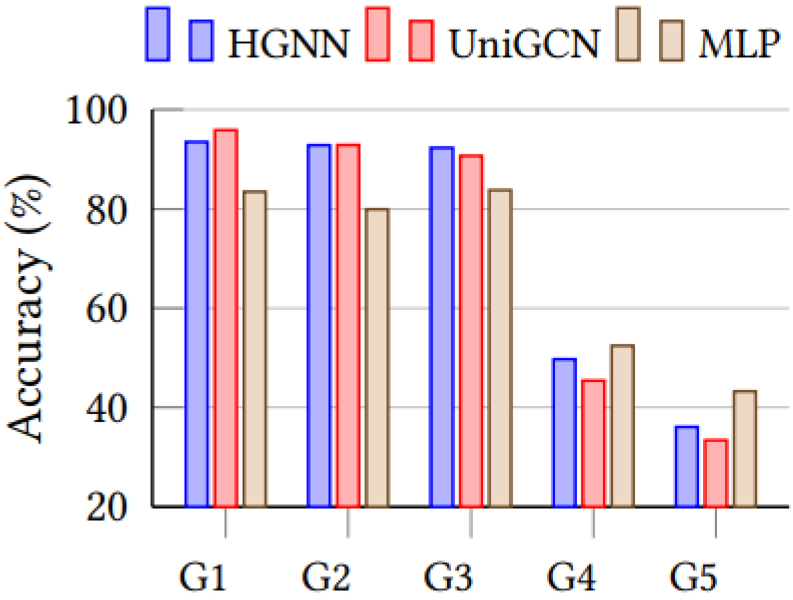
Hypernode Heterophily

$$h(v) = \sum_{e \in \mathcal{E}(v)} w_e \cdot h(e),$$

Observation of Heterophily

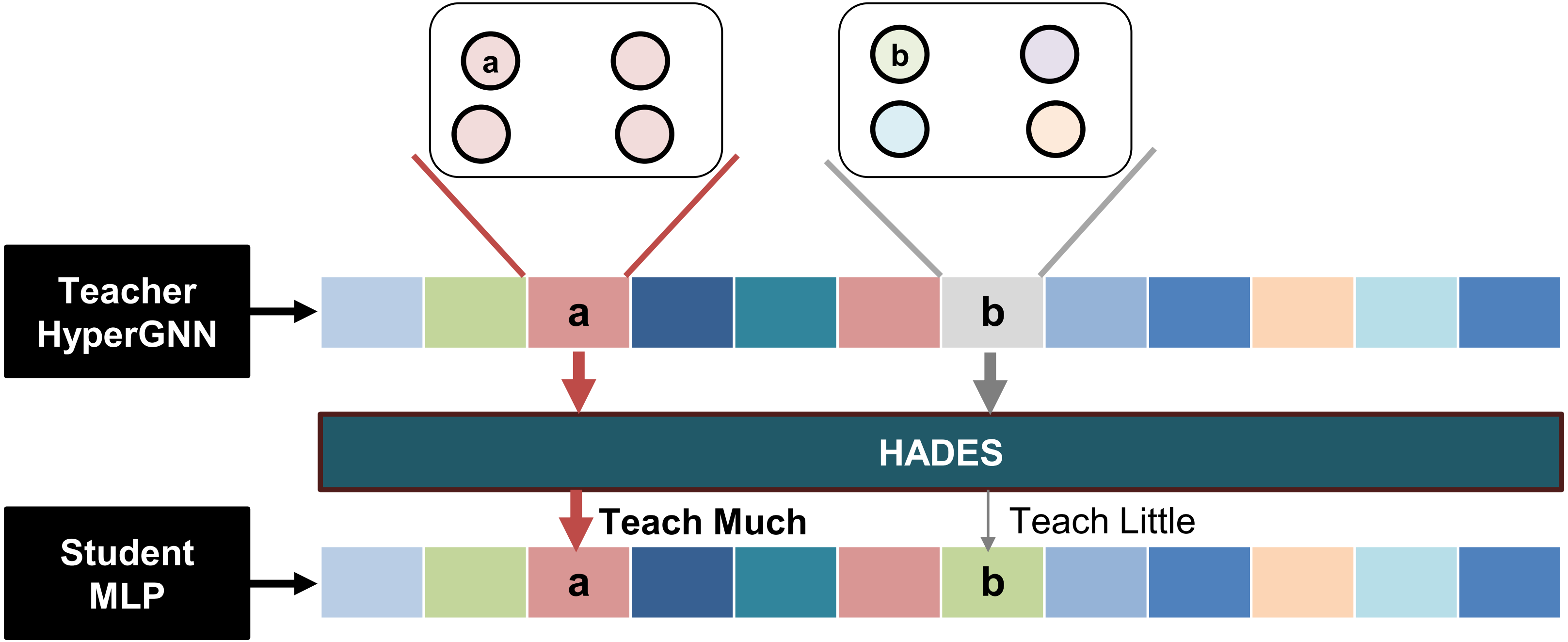
- In even homophily datasets, there are some heterophily nodes where MLP perform better than HyperGNN teachers





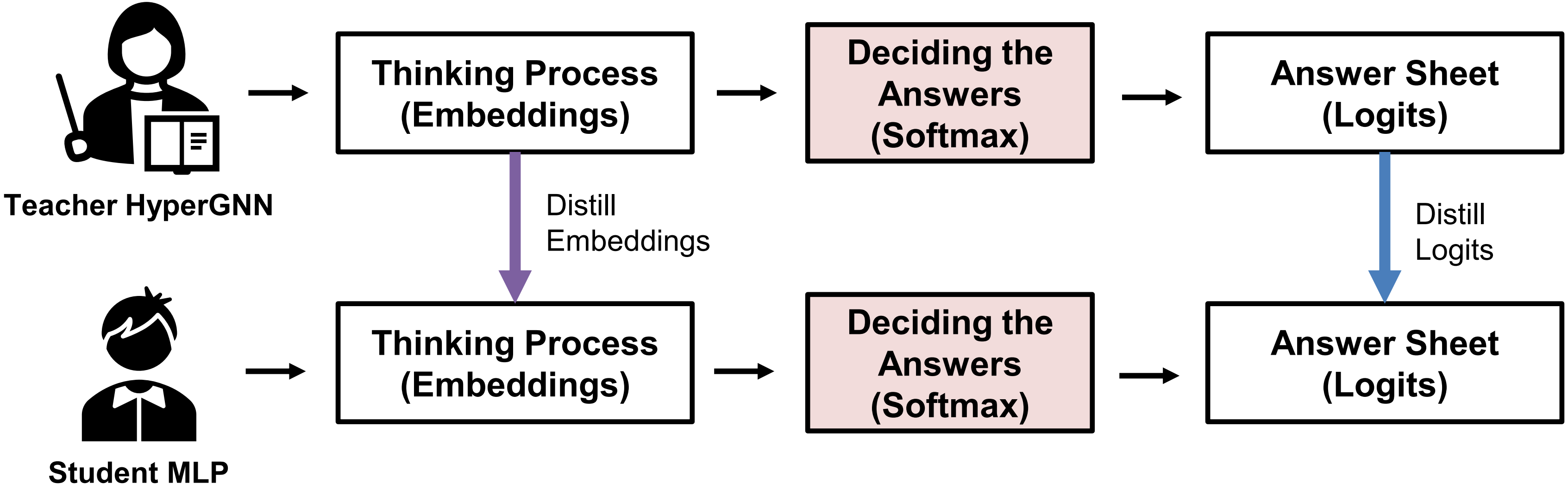
HADES

- Heterophily-aware AdaptivE DiStillation Method
- Consider node heterophily as a proxy for the reliability of teacher knowledge



What Can be Distilled?

- Most of the distillation techniques distill embeddings or logits
- If performance gain occurs in both embeddings and logits loss, it means that the method can be applied to most of the distillation methods



Heterophily-aware Distillation

- Apply heterophily-based scores as weights to distillation loss

Basic Distillation Loss	$\mathcal{L}_{Distill} = \lambda_L \cdot \mathcal{L}_L + \lambda_E \cdot \mathcal{L}_E,$
Heterophily-aware Distillation Loss	$r(v) = \exp(-\beta h(v)),$ $\mathcal{L}^*_{Distill} = \frac{\sum_{v \in V} r(v) \cdot (\lambda_L \mathcal{L}_L(v) + \lambda_E \mathcal{L}_E(v))}{\sum_{v' \in V} r(v')}$
Total Loss	$\mathcal{L} = \mathcal{L}_{CE} + \alpha \cdot \mathcal{L}_{Distill}$

Experiment Setup

- ☐ Node classification task to predict class labels of unseen nodes
- ☐ 50% for training, 25% for validation, 25% for test

Table 1: Statistics of real-world hypergraphs.

Datasets	$\lvert\mathcal{V}\rvert$	$\lvert\mathcal{E}\rvert$	$\lvert\mathcal{C}\rvert$	$\lvert\mathcal{F}\rvert$	Hyperedge size dist. (min/Q1/Q2/Q3/max)
Citeseer	1,458	1,079	6	3,703	2/2/2/4/26
Cora	1,434	1,579	7	1,433	2/2/3/4/5
Pubmed	3,840	7,963	3	500	2/2/3/4/171
DBLP	41,302	22,363	6	4,543	2/2/3/5/202

Distillation Performance

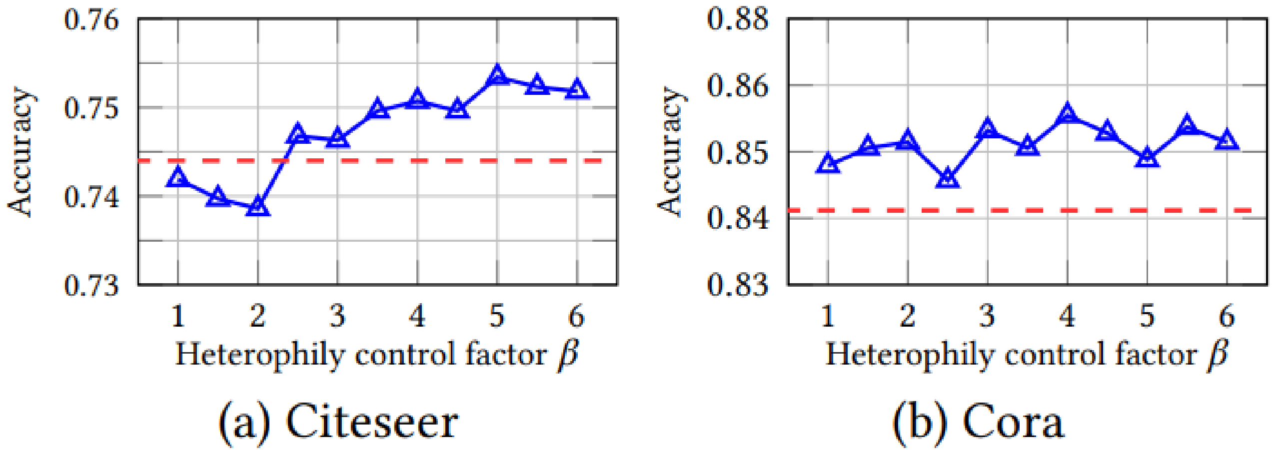
- HADES outperforms the baseline performance for most of the cases

Distilled	Datasets / Teacher	Citeseer T1: HGNN (0.733)	Citeseer T2: UniGCN (0.721)	Cora T1: HGNN (0.832)	Cora T2: UniGCN (0.825)	Pubmed T1: HGNN (0.849)	Pubmed T2: UniGCN (0.852)	DBLP T1: HGNN (0.906)	DBLP T2: UniGCN (0.905)
L	Baseline	0.744 ± 0.014	0.730 ± 0.010	0.839 ± 0.012	0.826 ± 0.018	0.866 ± 0.008	0.859 ± 0.010	0.914 ± 0.003	0.915 ± 0.002
	HADES	**0.753** ± 0.019	**0.747** ± 0.014	**0.858** ± 0.005	**0.836** ± 0.014	**0.886** ± 0.005	**0.878** ± 0.007	**0.923** ± 0.003	**0.925** ± 0.003
	Gain	+1.21%	+2.33%	+2.26%	+1.21%	+2.31%	+2.21%	+2.11%	+1.10%
E	Baseline	0.736 ± 0.020	0.729 ± 0.019	0.831 ± 0.005	0.767 ± 0.085	0.860 ± 0.007	0.803 ± 0.019	0.873 ± 0.005	0.855 ± 0.005
	HADES	**0.747** ± 0.015	**0.750** ± 0.018	**0.836** ± 0.012	**0.826** ± 0.020	**0.871** ± 0.007	**0.815** ± 0.013	**0.873** ± 0.005	**0.855** ± 0.003
	Gain	+1.49%	+2.88%	+0.60%	+7.69%	+1.28%	+1.49%	+0.00%	+0.00%
L+E	Baseline	0.745 ± 0.019	0.733 ± 0.010	0.827 ± 0.016	0.815 ± 0.018	0.861 ± 0.006	0.868 ± 0.007	0.913 ± 0.003	0.914 ± 0.003
	HADES	**0.752** ± 0.017	**0.740** ± 0.012	**0.863** ± 0.005	**0.836** ± 0.011	**0.882** ± 0.003	**0.875** ± 0.005	**0.923** ± 0.003	**0.923** ± 0.002
	Gain	+0.94%	+0.95%	+4.35%	+2.58%	+2.44%	+0.81%	+1.10%	+0.98%

Sensitivity Analysis

- Shows that suppressing potentially unreliable teacher knowledge from heterophilic nodes is beneficial
- Insensitive to the choice of β

$$r(v) = \exp(-\beta h(v)),$$



Inference Time Analysis

- When comparing between the distilled model using HADES and the teacher model (3-layers each), HADES is maximum 12.3x faster

Table 3: Inference time of two teacher models and HADES.

	Citeseer	Cora	Pubmed	DBLP
T1: HGNN	0.56 (ms)	0.54 (ms)	0.94 (ms)	5.27 (ms)
T2: UniGCN	1.16 (ms)	0.58 (ms)	1.84 (ms)	14.70 (ms)
HADES	**0.18** (ms) (3.1× / 6.4×)	**0.12** (ms) (4.5× / 4.8×)	**0.15** (ms) (6.3× / 12.3×)	**1.50** (ms) (3.5× / 9.8×)

Weighting Strategies for w_e

- Assign larger weights to larger hyperedges under the assumption that size of the hyperedges correlate to node heterophily

$$\boxed{h(v) = \sum_{e \in \mathcal{E}(v)} w_e \cdot h(e),}$$

(1) *Uniform*: $w_e = \frac{1}{|\mathcal{E}(v)|}$

(2) *Weighted*: $w_e = \frac{|e|}{\sum_{e' \in \mathcal{E}(v)} |e'|}$

(3) *Bound-Weighted*: $w_e = \frac{\min(|e|, Q_3)}{\sum_{e' \in \mathcal{E}(v)} \min(|e'|, Q_3)}$

(4) *Log-Weighted*: $w_e = \frac{\log(|e|+1)}{\sum_{e' \in \mathcal{E}(v)} \log(|e'|+1)}$

Weighting Strategies for w_e

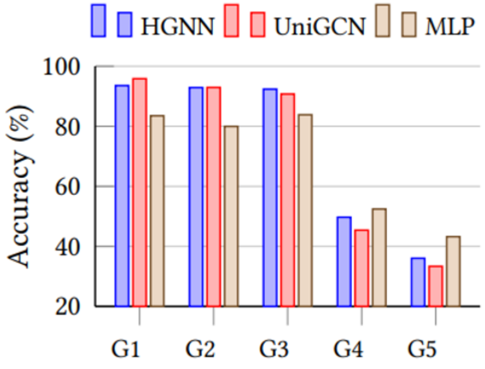
- In some cases, using the size of the hyperedges to estimate heterophily helps more effective knowledge transfer

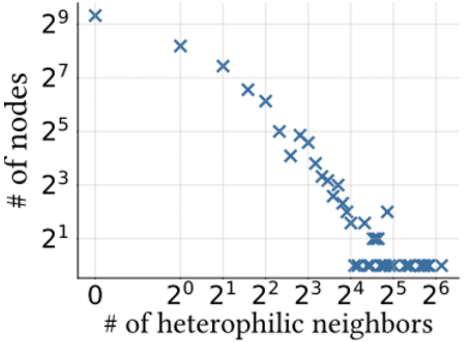
Table 4: Effect of heterophily quantification strategies.

Distilled	Method	Citeseer	Cora	Pubmed	DBLP	Rank ↓
L	Baseline	0.744	0.839	0.866	0.914	5.00
	Uniform	**0.753**	0.858	**0.886**	**0.923**	**1.75**
	Weighted	0.748	0.861	0.880	0.922	3.00
	Bound-Weighted	0.751	0.859	0.882	0.922	2.25
	Log-Weighted	0.750	**0.864**	0.882	0.919	2.50
E	Baseline	0.736	0.831	0.860	0.873	3.25
	Uniform	0.747	0.836	**0.871**	**0.873**	**2.25**
	Weighted	0.746	**0.849**	0.817	0.859	3.75
	Bound-Weighted	**0.752**	0.847	0.842	0.860	2.50
	Log-Weighted	0.751	0.847	0.828	0.863	2.75
L+E	Baseline	0.745	0.827	0.861	0.913	5.00
	Uniform	**0.752**	0.863	**0.882**	**0.923**	1.75
	Weighted	0.751	0.863	0.876	0.922	3.25
	Bound-Weighted	**0.752**	**0.864**	0.879	**0.923**	**1.50**
	Log-Weighted	**0.752**	0.861	0.880	**0.923**	2.00

Contributions

- ☐ Observed that HyperGNN distillations suffer **performance degradation on heterophilic nodes**, even in homophilic datasets
- ☐ Proposed a novel hypergraph distillation method, **HADES**, which uses heterophily scores for effective distillation
- ☐ Performance improvement was made in many datasets; however, **inference time did not increase** compared to previous distillation methods
- ☐ Can be **applied to any previous distillation** method





Thank You!